

REALTECH STP

Simple MQTT and SFTP Guide

Complete MQTT device details and application usability updates

Version	Trip topic	Main update
Application v0.1.42	IWS_STP-001/Device_Response	Server status, pending clear and startup loading

**What changed**

Every MQTT response sent by the device now identifies the site, device and lane using siteId, deviceId, laneId and deviceName. Full details are also shown in normal, exceptional and recharge examples.

Prepared for operations, server integration, QA and commissioning teams.

1. Common device details in every response

The server can identify exactly which STP site, controller and lane sent a message. These four fields are included in every device-to-server MQTT response.

siteld Manurankulam STP Yard Identifies the site or yard.	deviceId IWS STP 001 Identifies the physical controller.	laneId Manurankulam Tambaram Identifies the gate or lane.	deviceName STP-IWS-001 Human-readable device name.
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Common identity block

The same four values appear in normal IN, normal OUT, Manual exception, Auto exception, Applied recharge, Rejected recharge and Duplicate recharge responses.

Messages at a glance

Purpose	Direction	Topic / Interface	Plain meaning
Vehicle trip	Device -> Server	IWS_STP-001/Device_Response	IN, OUT or approved exception trip.
Recharge request	Server -> Device	IWS_STP-001/STP_RECHARGE_FROM_SERVER	Add balance to one RFID.
Recharge result	Device -> Server	IWS_STP-001/STP_RECHARGE_ACK_FROM_DEVICE	Applied, Duplicate or Rejected.
Image upload	Device -> SFTP	Configured remote SFTP folder	Upload the image before trip JSON.

2. Direction and normal processing

Direction is detected automatically from the RFID reader that reads the tag.

What happened	direction	What the device does
IN RFID reader detects the tag	IN	Use the IN camera and send an IN trip.
OUT RFID reader detects the tag	OUT	Use the OUT camera and send an OUT trip.



Reader rule

IN reader detection produces direction IN. OUT reader detection produces direction OUT.

Normal and approved-exception process



- A Paid IN trip deducts the configured amount from the local RFID balance.
- An OUT trip normally links to the earlier IN transaction and does not deduct again.
- A Manual or Auto approved exception uses the same camera, SFTP and MQTT process as the current IN or OUT trip.
- If SFTP or MQTT is unavailable, the transaction remains pending and is retried later.

3. Normal vehicle response examples

IN reader response

The IN RFID reader detected the tag. The payload contains complete device, vehicle, balance, image and time details.

```
{
  "topic": "IWS_STP-001/Device_Response",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.1",
    "transactionId": "2663656e-1fa6-4696-b93c-3f46662572aa",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS STP 001",
    "laneId": "Manurankulam Tambaram",
    "deviceName": "STP-IWS-001",
    "direction": "IN",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "accessType": "Paid",
    "previousBalance": 9800,
    "debitAmount": 100,
    "newBalance": 9700,
    "imageCaptureStatus": "Captured",
    "imageFileName": "...10-33-52 - IN.png",
    "imageRemotePath": "/home/.../STP-IWS-001/...IN.png",
    "tripDate": "2026-07-27",
    "tripTime": "10:33:52.823",
    "timeZoneOffset": "+05:30",
    "processedAt": "2026-07-27T10:33:52.823+05:30"
  }
}
```

4. Normal OUT response

The OUT RFID reader detected the tag. The same device details are sent, and entryTransactionId links this OUT event to the earlier IN event.

```
{
  "topic": "IWS_STP-001/Device_Response",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.1",
    "transactionId": "9a3b5b9e-9c87-420d-abcb-bd2f5f7b8954",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS STP 001",
    "laneId": "Manurankulam Tambaram",
    "deviceName": "STP-IWS-001",
    "direction": "OUT",
    "entryTransactionId": "2663656e-1fa6-4696-b93c-3f46662572aa",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "accessType": "Paid",
    "previousBalance": 9700,
    "debitAmount": 0,
    "newBalance": 9700,
    "imageCaptureStatus": "Captured",
    "imageFileName": "...11-20-44 - OUT.png",
    "imageRemotePath": "/home/.../STP-IWS-001/...OUT.png",
    "tripDate": "2026-07-27",
    "tripTime": "11:20:44.310",
    "timeZoneOffset": "+05:30",
    "processedAt": "2026-07-27T11:20:44.310+05:30"
  }
}
```

i

Same identity in both directions

Only the trip-specific values change. The site, device, lane and device name remain present in both IN and OUT messages.

5. Manual exceptional approval response

The OUT reader detected the RFID. After Manual approval, the OUT camera captures and uploads the image exactly like a normal OUT trip. Approval details are included in the full response.

```
{
  "topic": "IWS_STP-001/Device_Response",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.1",
    "transactionId": "46bc7046-530c-4242-9913-87b2d19f61d0",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS STP 001",
    "laneId": "Manurankulam Tambaram",
    "deviceName": "STP-IWS-001",
    "direction": "OUT",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "accessType": "Paid",
    "previousBalance": 10000,
    "debitAmount": 0,
    "newBalance": 10000,
    "imageCaptureStatus": "Captured",
    "imageFileName": "...12-10-05 - OUT.png",
    "imageRemotePath": "/home/.../STP-IWS-001/...OUT.png",
    "exceptionalApproval": {
      "approvalMode": "Manual",
      "reason": "Missing IN Trip",
      "approverName": "Arun Kumar",
      "approverRole": "Shift Supervisor",
      "approverMobile": "9876543210",
      "approvalDate": "2026-07-27",
      "approvalTime": "12:09:58.900",
      "timeZoneOffset": "+05:30"
    }
  }
}
```

6. Auto exceptional approval response

This is a complete Auto-approved IN response, not only the approval section. It carries the same device details and normal image information.

```
{
  "topic": "IWS_STP-001/Device_Response",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.1",
    "transactionId": "b115d46c-ea44-41b3-ba4c-ca20cca3b082",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS STP 001",
    "laneId": "Manurankulam Tambaram",
    "deviceName": "STP-IWS-001",
    "direction": "IN",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "accessType": "Paid",
    "previousBalance": 10000,
    "debitAmount": 100,
    "newBalance": 9900,
    "imageCaptureStatus": "Captured",
    "imageFileName": "...12-30-10 - IN.png",
    "imageRemotePath": "/home/.../STP-IWS-001/...IN.png",
    "exceptionalApproval": {
      "approvalMode": "Auto",
      "reason": "Missing OUT Trip",
      "approverName": "SYSTEM",
      "approverRole": "IN AUTO APPROVAL",
      "approverMobile": "N/A",
      "approvalDate": "2026-07-27",
      "approvalTime": "12:30:10.200",
      "timeZoneOffset": "+05:30"
    }
  }
}
```

i

Approval identification

approvalMode clearly identifies Manual or Auto. Name, role and mobile fields are still present for both modes.

7. Recharge request and Applied response

Server sends the request

```
{
  "topic": "IWS_STP-001/STP_RECHARGE_FROM_SERVER",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.0",
    "messageType": "rfidRecharge",
    "rechargeId": "RCH-20260727-000001",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS STP 001",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "previousBalance": 9700,
    "rechargeAmount": 300,
    "newBalance": 10000
  }
}
```

Device sends Applied response

```
{
  "topic": "IWS_STP-001/STP_RECHARGE_ACK_FROM_DEVICE",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.0",
    "messageType": "rfidRechargeAck",
    "rechargeId": "RCH-20260727-000001",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS STP 001",
    "laneId": "Manurankulam Tambaram",
    "deviceName": "STP-IWS-001",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "status": "Applied",
    "success": true,
    "duplicate": false,
    "rechargeAmount": 300,
    "previousBalance": 9700,
    "newBalance": 10000,
    "serverExpectedNewBalance": 10000,
    "serverBalanceMatched": true,
    "processedAt": "2026-07-27T10:58:19.604+05:30",
    "message": "Recharge applied to the local RFID balance."
  }
}
```


8. Rejected and Duplicate recharge responses

Both responses include the same complete device identity, allowing the server to identify the exact source even when the recharge was not newly applied.

Rejected

```
{
  "topic": "IWS_STP-001/STP_RECHARGE_ACK_FROM_DEVICE",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.0",
    "messageType": "rfidRechargeAck",
    "rechargeId": "RCH-20260727-000002",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS_STP_001",
    "laneId": "Manurankulam Tambaram",
    "deviceName": "STP-IWS-001",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "status": "Rejected",
    "success": false,
    "duplicate": false,
    "rechargeAmount": 300,
    "serverExpectedNewBalance": 10000,
    "processedAt": "2026-07-27T10:57:38.315+05:30",
    "message": "Recharge request validation failed."
  }
}
```

Duplicate

```
{
  "topic": "IWS_STP-001/STP_RECHARGE_ACK_FROM_DEVICE",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.0",
    "messageType": "rfidRechargeAck",
    "rechargeId": "RCH-20260727-000001",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS_STP_001",
    "laneId": "Manurankulam Tambaram",
    "deviceName": "STP-IWS-001",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "status": "Duplicate",
    "success": true,
    "duplicate": true,
    "rechargeAmount": 300,
    "previousBalance": 9700,
    "newBalance": 10000,
    "serverExpectedNewBalance": 10000,
    "serverBalanceMatched": true,
    "processedAt": "2026-07-27T11:00:00.000+05:30",
    "message": "Recharge was already processed."
  }
}
```

9. SFTP upload and final checklist

SFTP moves the captured IN, OUT or approved-exception image to the server folder. MQTT then carries the uploaded image path and complete device details.



SFTP remains separate

SFTP uploads the image file. MQTT sends the JSON containing siteld, deviceId, laneId, deviceName and imageRemotePath.

Practical checklist

- 1 Use IWS_STP-001/Device_Response for every vehicle trip response.
- 2 Include siteld, deviceId, laneId and deviceName in every device-to-server MQTT response.
- 3 IN RFID reader detection must produce direction IN.
- 4 OUT RFID reader detection must produce direction OUT.
- 5 Normal and approved-exception trips must capture the correct lane camera image.
- 6 Upload the image through SFTP before publishing the trip JSON.
- 7 Include full device details in Manual and Auto exceptional approval payloads.
- 8 Include full device details in Applied, Rejected and Duplicate recharge acknowledgements.
- 9 Use approvalMode, reason, approverName, approverRole and approverMobile for exceptions.
- 10 Store transactionId and rechargeId to prevent duplicate processing.



Final understanding

Every response tells the server what happened and exactly which site, device and lane sent it.

10. Application usability updates

The main application now gives the operator clearer information during startup and daily server synchronization.

Internet status Shows whether the computer has network or internet access.	Server status Shown next to Internet. Indicates MQTT and SFTP reachability.	Pending data Data > Clear Pending Data permanently removes selected pending records after confirmation.
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Server connectivity status

- Green: MQTT and the enabled SFTP endpoint are reachable.
- Orange: only one server service is reachable, such as MQTT online but SFTP offline.
- Red: the configured server endpoints are offline or not configured.



Simple meaning

Internet status confirms general network access. Server status separately confirms whether this application can reach its configured MQTT and SFTP server ports.

Clear Pending Data



Permanent action

The application shows the number of pending transactions and asks for confirmation. Use the option only when no vehicle is being processed. Synced transaction history is not deleted.

Startup loading screen

A visible orange loading window opens immediately and reports each startup stage instead of leaving the operator waiting without feedback.



No artificial delay

The loading screen closes as soon as startup is complete. It does not intentionally slow the application.